The Theory of Dobble

John Longley, 7 November 2019

This handout contains some further information about the mathematical theory behind Dobble, following on from my talk.

I'm using the term '*Level N Dobble*' to refer to the version of Dobble with N symbols per card. So 'classic Dobble' is Level 8. In the talk, we started by constructing versions of Dobble at Levels 2 and 3 (Level 1 is extremely boring!), and moved on to see how Level 8 worked.

The standard Dobble rule is 'Any two cards share exactly one symbol'. We are also calling a set of Dobble cards *complete* if, in addition, it's true that *any two symbols share exactly one card*. The Level 2 and 3 sets that we constructed were complete sets, as was the 57-card version of Level 8 Dobble that we arrived at later on.

I showed you the following table giving the sizes of complete Level N Dobble sets for various numbers N:

Level	Number of cards = Number of symbols
1	1
2	3
3	7
4	13
5	21

(As a puzzle, you might enjoy trying to construct the Level 4 and Level 5 systems – e.g. using the 13 letters A, ..., M and the 21 letters A, ..., U. *Warning:* the Level 5 one is slightly tricky!)

This table raises two obvious questions:

1. Does a complete Level N Dobble set exist for *every* whole number N?
2. What is the pattern here? Is there a formula for the number of cards/symbols in a complete Level N Dobble set (if such a set exists)?

I'll say a bit about question 1 first, without going into details. Complete Level N sets do exist for many numbers N, but not for all. For example, it's known that *there is no Complete Dobble set* at level 7 or 11. (This was shown by getting a computer to try out all possibilities.) Other than that, complete Dobble sets do exist for all levels up to 12. Intriguingly, for level 13, it is currently an *open question* whether a complete set exists – no one knows the answer yet! (In this case, trying out all possibilities is beyond the reach of today's computers.)

Now for question 2. You might be able to spot a pattern in the above table: taking the differences between successive numbers, we get 2,4,6,8, so the 'difference of differences' is always 2. In fact, it turns out that in a complete Level N Dobble set (if one exists), both the number of cards and the number of symbols must be exactly

$$N^2 - N + 1$$

(You can check that this agrees with the numbers in the table.) What's more, it's actually not too hard to *prove* this fact, and this is what I'd like to do over the next three pages.

So... let's suppose N is a whole number (we shall assume $N \geq 2$ from now on), and let's suppose we had a complete Dobble set of level N (i.e. N symbols per card). Notice that we're not saying there *is* such a set – we've already seen that complete Dobble sets don't exist for some numbers N – but even so, we can ask what would be true about such a set if it *did* exist. We'll also suppose that the set we're talking about contains at least two cards (otherwise things are boring!).

We'll now prove a series of facts that would necessarily be true for a Dobble set of this kind. In some cases, the proofs of these facts rely on earlier facts, so that some of our facts rest on others, like the blocks in a building. Working through all these proofs might take quite a bit of time and patience, but they will give you a good feel for what mathematical proofs are like.

FACT 1: *The number of cards is at least* $(N-1)^2 + 2$.

Proof: Take any two cards: call them A and B. These will have just one symbol in common: call it s. Card A will then contain $N-1$ other symbols beside s: call them $a_1, a_2, ..., a_{N-1}$. Likewise, Card B will contain $N-1$ other symbols beside s: call them $b_1, b_2, ..., b_{N-1}$. Notice that none of the symbols $a_1, a_2, ..., a_{N-1}$ can be the same as any of the symbols $b_1, b_2, ..., b_{N-1}$: otherwise the cards A and B would have another symbol in common beside s, which would contradict the rule that any two cards have exactly one symbol in common.

There are now $(N-1) \times (N-1) = (N-1)^2$ ways in which it's possible to choose a symbol a from $a_1, a_2, ..., a_{N-1}$ together with a symbol b from $b_1, b_2, ..., b_{N-1}$. What's more, for any such choice of a and b, there must be some card containing both a and b, since we're assuming our set is complete. Call this card C_{ab} (it will depend on what a and b are). Notice that C_{ab} can't be the same card as A, because we've already seen that A doesn't contain any of the b symbols. Likewise, C_{ab} can't be B, because B doesn't contain any of the a symbols.

The last step is to show that each of our $(N-1)^2$ possible pairs a,b gives rise to a different card C_{ab}. To see this, suppose a,b and a',b' are two different pairs: we want to show that C_{ab} and $C_{a'b'}$ can't be the same card. First, note that we can't have both $a = a'$ and $b = b'$: otherwise a,b and a',b' would be exactly the same pair! So either $a \neq a'$ or $b \neq b'$ (or both). Let's consider these two situations in turn. In both cases, we want to show that C_{ab} and $C_{a'b'}$ can't be the same card.

Suppose $a \neq a'$. Then if C_{ab} and $C_{a'b'}$ were the same card, this card would contain both a and a'. But we already have another card containing both a and a', namely A! So we have more than one card containing a and a' ; but this contradicts the completeness rule which says that any two symbols share *exactly one* card.

Now suppose $b \neq b'$. In this case, a precisely similar argument applies: if if C_{ab} and $C_{a'b'}$ were the same card, then this card would contain both b and b'. But we already have another card that does that, namely B. So once again, the completeness rule is contradicted.

Summing up, we see that we have a card C_{ab} for each of our $(N-1)^2$ possible pairs a,b, plus the two cards A and B we started with. What's more, we've seen that *all of these cards are different*! So we've shown that there must be *at least* $(N-1)^2 + 2$ cards (there may of course be others). ■

FACT 2: *There is a symbol that appears on at least N cards.*

Proof: Pick any card A. It has N symbols on it. Each of the other cards must contain exactly one of these N symbols, since it has to share a symbol with A. So we can sort all of the other cards into N piles, according to the symbols they have in common with A.

Now, one of these piles must contain at least $N-1$ cards: if they all contained $N-2$ cards or fewer, then we would have at most $N \times (N-2) + 1$ cards in total, including A itself. But Fact 1 tells us we have at least $(N-1)^2 + 2$ cards, which is more than $N \times (N-2) + 1$:

$$(N-1)^2 + 2 \;=\; N^2 - 2N + 1 + 2 \;>\; N^2 - 2N + 1 \;=\; N \times (N-2) + 1.$$

So there's a pile with at least $N-1$ cards, all of which have the same symbol s in common with A. So including A itself, there are at least N cards containing the symbol s. This proves Fact 2. ■

FACT 3: *There are at least $N^2 - N + 1$ different symbols.*

Proof: Using Fact 2, choose a symbol s that appears on at least N cards: say on the cards A_1, A_2, ..., A_N. Each of these cards has $N-1$ symbols besides s, and what's more, *all of these symbols are different*. This is because if the same symbol t (different from s) appeared on two of these cards, then those two cards would have both s and t in common, contradicting the basic Dobble rule.

So on these N cards, we have the symbol s (appearing N times), and $N \times (N-1)$ other symbols, all different. That makes $N \times (N-1) + 1 = N^2 - N + 1$ different symbols (even without looking at any of the other cards). ■

FACT 4: *There's no symbol that appears on* all *the cards.*

Proof: Suppose there were a symbol s that appeared on every card. Take two cards A and B, both containing s. Now choose a symbol a from A, different from s, and a symbol b from B, also different from s. Clearly b cannot appear on A, since that would mean that A and B had two symbols in common (s and b). On the other hand, since we're assuming our set is complete, there must be some card C containing both a and b. Now C must be different from A, since C contains b and A doesn't. But if it's really true that s appears on every card, then s must appear on A, which means that C and A have both s and a in common. This contradicts the basic Dobble rule.

This means it can't be true after all that there's a symbol s that appears on every card. And this is what we wanted to show. ■

FACT 5: *There are* exactly *$N^2 - N + 1$ different symbols.*

Proof: We've already shown in Fact 3 that there are at least this many symbols, so we just need to show that there are no more than this.

Let's return to the situation we had in the proof of Fact 3, where we had N cards A_1, A_2, ..., A_N, all containing some symbol s, and between them containing $N \times (N-1)$ other symbols, all different. That already gives us $N^2 - N + 1$ different symbols, so we just need to show that there are no other symbols besides these on any of the other cards.

Suppose there were a card C that contained a new symbol t, not appearing on any of A_1, A_2, ..., A_N. Then either C contains the symbol s, or it doesn't. We will show that in either of these cases, we are led to a contradiction.

Case 1: C doesn't contain s. Note that C must have a symbol in common with A_1 (call it a_1), and a symbol in common with A_2 (call it a_2), and so on for all the others. This gives us symbols $a_1, a_2, ..., a_N$ which must all appear on C. But since s doesn't appear on C, these symbols are all different from s, and what's more, they are all different from each other, since no two of the cards $A_1, A_2, ..., A_N$ can have another symbol in common besides s. So we have N *different* symbols $a_1, a_2, ..., a_N$ which must all appear on C. But this leaves no room for the new symbol t which is meant to appear on C. Contradiction!

Case 2: C does contain s. This means that all the symbols on C other than s must be 'new', i.e. they don't appear anywhere on $A_1, A_2, ..., A_N$, because none of these cards can have common symbol with C besides s.

Using Fact 4, pick a card D that *doesn't* contain s. Now C and D must have a symbol in common: call it u. (It may or may not be the same as t.) By what we've said above, u must be a new symbol not appearing on $A_1, A_2, ..., A_N$. We are now in the situation that we have a card D containing a new symbol u, where D doesn't contain s. So we can now use the argument from Case 1, but applying it to D and u rather than to C and t. This again leads us to a contradiction.

So there's no way out! We're led to a contradiction in either case, so we're forced to conclude that there cannot be a new symbol t after all. This means that the $N^2 - N + 1$ symbols that we know from the cards $A_1, A_2, ..., A_N$ are the only symbols there are. ■

FACT 6: *No symbol can occur on more than N cards.*

Proof: This is basically shown by arguments we've used already. Suppose some symbol s appeared on (at least) the $N + 1$ cards $A_1, A_2, ..., A_{N+1}$. Just as in the proof of Fact 3, we see that each of these cards contains $N - 1$ non-s symbols, and all of these are different, because no two of these cards can share another symbol besides s. So we have at least $((N + 1) \times (N - 1)) + 1 = N^2 - 1 + 1 = N^2$ different symbols. But this is impossible, since we've seen in Fact 5 that there are only $N^2 - N + 1$ different symbols. (Note that if $N > 1$ then $N^2 - N + 1 < N^2$.) ■

FACT 7: *Each symbol appears on* exactly *N cards.*

Proof: We've seen in Fact 6 that each symbol appears on at most N cards, so we just have to rule out the possibility that a symbol appears on fewer than N cards.

Suppose some symbol s appears on the cards $A_1, A_2, ..., A_m$, where $m < N$, and on no others. Each of these m cards has $N - 1$ symbols besides s, so we have just $m \times (N - 1) + 1$ different symbols on these cards. But this is less than $N \times (N - 1) + 1$, which is the total number of different symbols overall (Fact 5). So there must be some symbol t that doesn't appear on any of $A_1, A_2, ..., A_m$, but does appear on some other card.

By the completeness property, there must be some card C containing both s and t. But this is a contradiction, since $A_1, A_2, ..., A_m$ are the only cards containing s, and none of these contain t. We've therefore shown that it's impossible for a symbol s to appear on fewer than N cards. ■

FACT 8: *There are exactly $N^2 - N + 1$ cards.*

Proof: By Fact 5, there are exactly $N^2 - N + 1$ different symbols, and by Fact 7, each of these appears exactly N times in total. So the total number of *symbol occurrences* in the entire set is at most $N \times (N^2 - N + 1)$. But the total number of symbol occurrences is also $N \times$ the number of cards. So the number of cards must also be exactly $N^2 - N + 1$. ■

Summing up

Taken together, the above facts give us a remarkably specific picture of what *any* complete Level N Dobble set has to look like, if it exists. There must be exactly $N^2 - N + 1$ cards and the same number of symbols; and not only does every card have N symbols, but every symbol must appear on exactly N cards.

Let's now summarize what we know. Here are the numbers of cards/symbols in complete level N Dobble sets for small values of N:

Level	Number of cards = Number of symbols
1	1
2	3
3	7
4	13
5	21
6	31
7	doesn't exist
8	57
9	73
10	91
11	doesn't exist
12	133
13	157 if it exists (not known!)

In the talk, I showed a method for constructing a complete Level 8 set with 57 cards/symbols. This method relied on the fact that $8 - 1 = 7$ is a *prime number*. (In terms of the geometry, we need this to ensure that there is only one line joining any two points.) In fact, the same method will work for *any* value of N that is one more than a prime number. So this takes care of the cases $N = 3, 4, 6, 8, 12$ in the above table (and also applies e.g. when $N = 14$). To construct complete sets for the cases $N = 5$, 9 and 10, we need slightly different methods, which I won't go into here. (As mentioned earlier, the case $N = 5$ is possible even without a specific method, although quite challenging.)

Reconstructing the magic 7 × 7 grid

Finally, if you have an ordinary Dobble set of your own, here are some instructions for how to re-create the 7 × 7 arrangement I showed in my talk, with each of the 56 lines identified by some symbol. It may take you around 10-15 minutes to do this.

First, find all 6 cards containing *the snowman*, and set them aside. (These will all be 'direction cards'. There would of course be 8 snowman cards in a *complete* set, but in the commercial version of Dobble, two are missing.) The remaining 49 cards will be the ones you will arrange in a grid.

Find a surface big enough for a 7 × 7 grid of Dobble cards, and *imagine* that the rows and columns are labelled with symbols like this:

	excl-mark	ladybird	hammer	dog	bones	light-bulb	eye
ice-cube							
daisy							
gingerbread man							
dinosaur							
cactus							
maple-leaf							
question mark							

Now go through each of the 49 non-snowman cards in turn. Each of them will contain exactly one of the row-symbols and exactly one of the column-symbols, and these will tell you where to place the card in the grid. For instance, if your card contains the ice-cube and the eye, it will go in the top-right corner of the grid. If it contains the dinosaur and the dog, it will go right in the middle. If it contains the question mark and the exclamation mark, it will go bottom-left.

You will find that this fills up the grid with each of the 49 cards occupying a different square. Now step back and admire all the lines you can pick out – and amaze your friends!

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